

Fused Acoustic and Text Encoding for Multimodal Bilingual Pretraining and Speech Translation

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Introduction

Existing representation learning models suffer from two limitations:

1. They learn from only one input modality, and thus, do not have a **unified representation**
2. Their performance is limited by the **scarcity** of parallel speech translation **data**

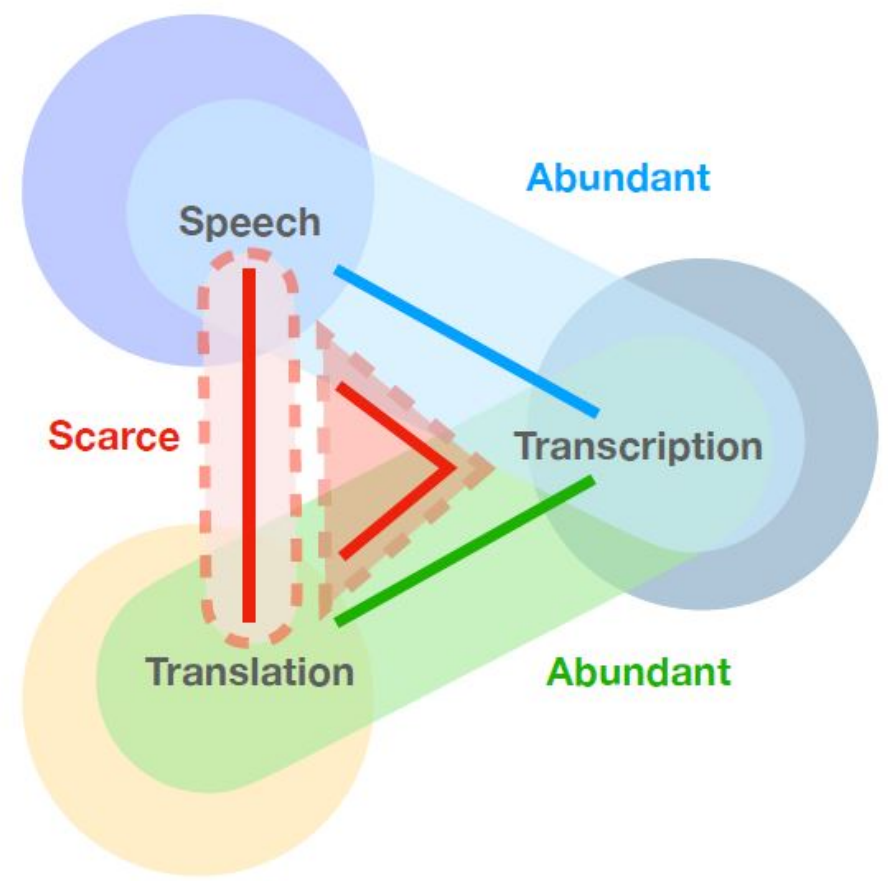


Figure 1. Though speech translation datasets are scarce, there is an abundance of speech, text and speech recognition datasets that can be leveraged

Main Contributions

1. Propose two models **FAT-MLM**, which can learn a unified acoustic and text representation; and **FAT-ST**, which can do speech recognition and machine translation in a single encoder-decoder framework
2. Achieve **SOTA** for end-to-end speech translation, while maintaining a smaller model size and faster decoding time

Fused Acoustic and Text Masked Language Model (FAT-MLM)

- Existing pre-training models can learn the representations for either text or speech
- FAT-MLM is a **multimodal** pretraining model which encodes acoustic and textual information into a **unified** representation

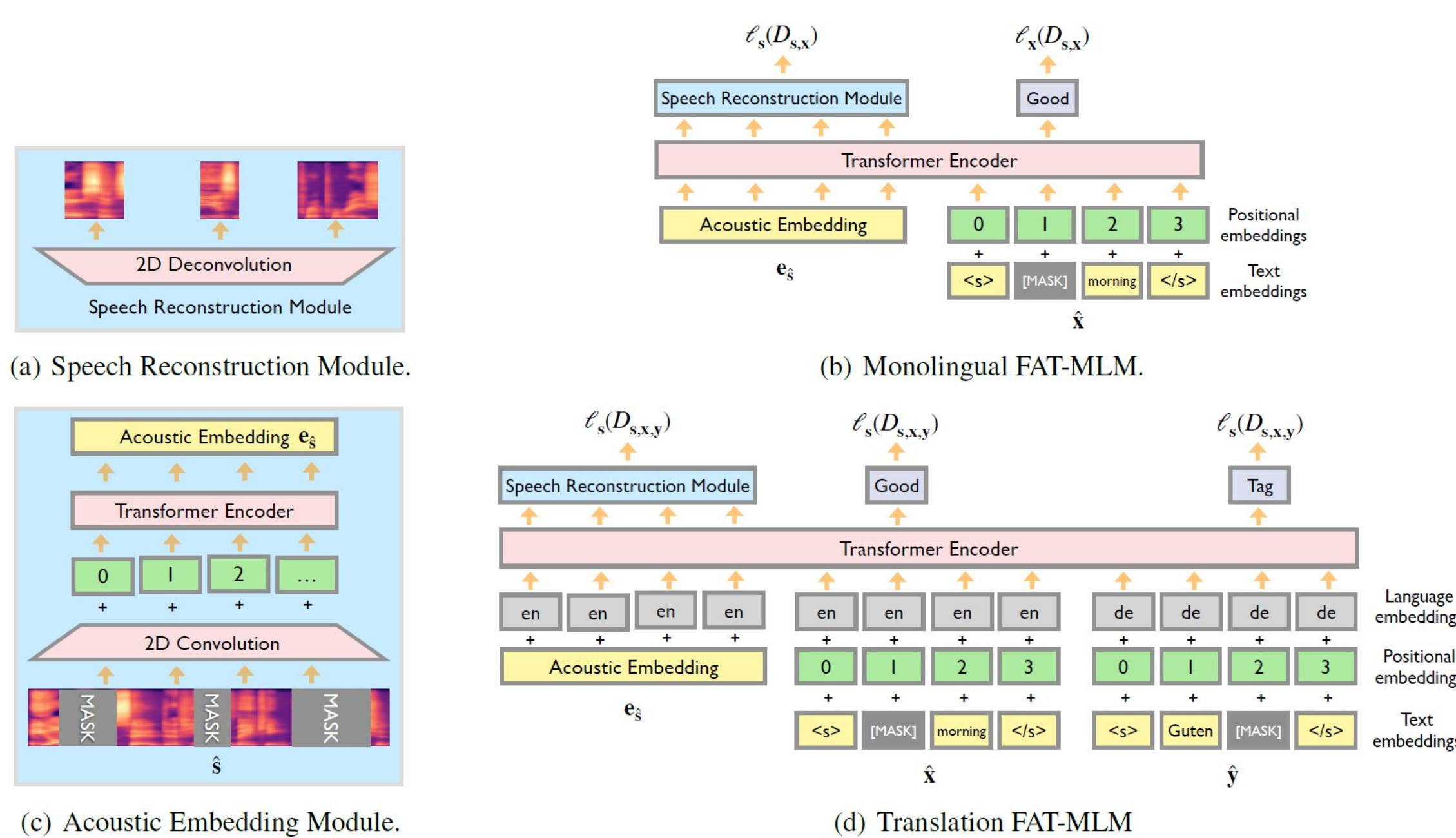


Figure 3. Fused Acoustic and Text-Masked Language Model (FAT-MLM).

Monolingual FAT-MLM

1. Input $D_{s,x} = \{(\mathbf{s}, \mathbf{x})\}$, where $\mathbf{s} = (s_1, \dots, s_{|\mathbf{s}|})$ is the sequence of acoustic features $s_i \in \mathbb{R}^d$ and $\mathbf{x} = (x_1, \dots, x_{|\mathbf{x}|})$ is the sequence of corresponding transcriptions
2. Apply random masking function $\hat{s} \sim \text{Mask}_{\text{span}}(\mathbf{s}, \lambda)$ by replacing several random spans of s with probability λ with random vector $\varepsilon_s \in \mathbb{R}^{d_s}$
3. Create acoustic embeddings e_s by embedding $\hat{s}$ with Convolutions and a Transformer encoder
4. As for s , also randomly mask tokens in $\mathbf{x}$ via $\hat{x} \sim \text{Mask}_{\text{token}}(\mathbf{x}, \lambda)$
5. Concatenate acoustic and source text embeddings to get $[\hat{e}_s; \hat{\mathbf{x}}]$ and encode using Transformer f to get the latent representations $f([\hat{e}_s; \hat{\mathbf{x}}])$
6. The training loss function is given by $l_{\text{FAT-MLM}}(D_{s,x}) = l_s(D_{s,x}) + l_x(D_{s,x})$
7. The Speech Reconstruction Loss is $l_s(D_{s,x}) = \sum_{(\mathbf{s}, \mathbf{x}) \in D_{s,x}} \|\mathbf{s} - g(f([\hat{e}_s; \hat{\mathbf{x}}]))\|_2^2$
8. The Text Reconstruction Loss is $l_x(D_{s,x}) = -\sum_{(\mathbf{s}, \mathbf{x}) \in D_{s,x}} \log p(\mathbf{x} | [\hat{e}_s; \hat{\mathbf{x}}])$

Translation FAT-MLM

1. For multimodal cross-lingual tasks such as speech translation. Input is $D_{s,x,y} = \{(\mathbf{s}, \mathbf{x}, \mathbf{y})\}$, where $\mathbf{y} = [y_1, \dots, y_{|\mathbf{y}|}]$ is the sequence of target language translation
2. Compute source and target language embeddings $\mathbf{e}_{src}$ and $\mathbf{e}_{tgt}$
3. Perform random masking on $\mathbf{y}$, after which concatenate to get $\mathbf{h}_{s,x,y} = [\mathbf{e}_{\hat{s}} + \mathbf{e}_{src}; \hat{\mathbf{x}} + \mathbf{e}_{src}; \hat{\mathbf{y}} + \mathbf{e}_{tgt}]$
4. Pass concatenated embeddings $\mathbf{h}_{s,x,y}$ via Transformer encoder
5. Reconstruction losses are given by $l_s(D_{s,x,y}) = \sum_{(\mathbf{s}, \mathbf{x}, \mathbf{y}) \in D_{s,x,y}} \|\mathbf{s} - g(f(\mathbf{h}_{s,x,y}))\|_2^2$, $l_x(D_{s,x,y}) = -\sum_{(\mathbf{s}, \mathbf{x}, \mathbf{y}) \in D_{s,x,y}} \log p(\mathbf{x} | \mathbf{h}_{s,x,y})$ and $l_y(D_{s,x,y}) = -\sum_{(\mathbf{s}, \mathbf{x}, \mathbf{y}) \in D_{s,x,y}} \log p(\mathbf{y} | \mathbf{h}_{s,x,y})$. The final loss function is the sum of the three loss functions

Fused Acoustic and Text Speech Translation (FAT-ST)

Machine Translation to Speech Translation

- Recall that the loss function for Machine Translation is given by $p(\mathbf{y} | \mathbf{x})_{MT} = \prod_{t=1}^{|\mathbf{y}|} p(y_t | x, y_{<t})$ and the total loss is given by $\ell_{MT}(D_{x,y}) = -\sum_{(\mathbf{x}, \mathbf{y}) \in D_{x,y}} \log p(\mathbf{y} | \mathbf{x})$
- The loss for speech translation is similar to the MT loss and is given by $p(\mathbf{y} | \mathbf{s})_{ST} = \prod_{t=1}^{|\mathbf{y}|} p(y_t | s, y_{<t})$ and the total loss is given by $\ell_{ST}(D_{s,y}) = -\sum_{(\mathbf{s}, \mathbf{y}) \in D_{s,y}} \log p(\mathbf{y} | \mathbf{s})$

FAT-ST

- FAT-ST is an encoder-decoder architecture where the encoder is identical to the monolingual FAT-MLM architecture and is used to encode both acoustic or text data. The decoder is a standard transformer decoder
- We can then optimize this encoder by using speech translation loss $\ell_{ST}(D_{s,y})$, machine translation loss $\ell_{MT}(D_{s,x})$ and FAT-MLM loss $\ell_{\text{FAT-MLM}}(D_{x,y})$.
- The final loss is given by

$$\ell_{\text{FAT-ST}}(D_{s,y} \cup D_{s,x} \cup D_{x,y}) = \ell_{ST}(D_{s,y}) + \ell_{MT}(D_{s,x}) + \ell_{\text{FAT-MLM}}(D_{x,y})$$

Experiments

We conducted speech translation experiments in 3 directions: English to German (En→De), English to Spanish (En→Es), and English to Dutch (En→Ni) to show the translation quality of baselines and our proposed methods.

Pretrain Method	Models	En→De	En→Es	En→Ni	Avg.	Model Size
No Pretraining	ST	19.64	23.68	23.01	22.11	31.25M
	ST + ASR	21.70	26.83	25.44	24.66 (+2.55)	44.82M
	ST + ASR & MT	21.58	26.37	26.17	24.71 (+2.60)	56.81M
	ST + MAM	20.78	25.34	24.46	23.53 (+1.42)	33.15M
	ST + MAM + ASR	22.41	26.89	26.49	25.26 (+3.15)	46.72M
	Liu et al. (2020b)	22.55	-	-	-	51.20M
	Le et al. (2020)	23.63	28.12	27.55	26.43 (+4.32)	83.79M
	Cascade ¹	23.65	28.68	27.91	26.75 (+4.64)	83.79M
	FAT-ST (base).	22.70	27.86	27.03	25.86 (+3.75)	39.34M
	ASR & MT	21.95	26.83	26.03	24.94 (+2.83)	31.25M
MAM	ST + ASR & MT	22.05	26.95	26.15	25.05 (+2.94)	56.81M
	FAT-ST (base)	22.29	27.21	26.26	25.25 (+3.14)	39.34M
FAT-MLM	FAT-ST (base)	23.68	28.61	27.84	26.71 (+4.60)	39.34M
	FAT-ST (big)	23.64	29.00	27.64	26.76 (+4.65)	58.25M

Figure 2. BLEU comparisons on Must-C test set between our proposed methods and other baselines over 3 translation directions

We can see that our proposed FAT-ST base model achieves the best results except Le et al. (2020) and the cascaded model, which have much higher number of parameter than our model

Pretrain Data	Pretrain Method	Train Data	Models	En→De	En→Es	En→Ni	Avg.
$\emptyset$	ASR & MT	$D_{s,x,y}$	ST	19.64	23.68	23.01	22.11
			Cascade ¹	23.65	28.68	27.91	26.75 (+4.64)
$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-MLM	$D_{s,x,y}$	FAT-ST (base)	23.26	27.16	26.15	25.17 (+3.06)
			FAT-ST (big)	22.73	27.99	27.12	25.95 (+3.84)
$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-MLM	$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-ST (base)	23.98	28.95	28.08	27.00 (+4.89)
			FAT-ST (big)	24.34	29.41	28.86	27.54 (+5.45)
$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-MLM	$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-ST (base)	24.02	29.25	28.28	27.18 (+5.07)
			FAT-ST (big)	24.58	30.10	29.36	28.01 (+5.90)
$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-MLM	$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-ST (base)	23.91	29.01	28.18	27.03 (+4.92)
			FAT-ST (big)	25.47	30.75	30.08	28.77 (+6.66)
$\emptyset$		$D_{s,x,y} \cup D_{x,y}$	Pino et al. (2020)	25.2			

Figure 3. BLEU comparisons on Must-C test set between our proposed methods using additional data

Above table shows that FAT-MLM can further improve FAT-ST by simply adding speech recognition data $D_{s,x}$ (Librispeech), text machine translation data $D_{x,y}$ (Europarl) and even speech only data D_s (Libri-light) and monolingual text data $D_x \cup D_y$

Train Data	Pretrain Data	Models	→De	→Es	→Ni
$D_{s,x,y}$	No pretraining	MT ³	27.63	32.61	32.08
		FAT-ST (base)	24.41	30.81	29.18
	$D_{s,x,y}$	FAT-ST (base)	27.24	31.98	31.27
		FAT-ST (big)	26.92	32.29	31.48
	$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-ST (base)	27.43	32.38	32.44
		FAT-ST (big)	27.60	32.95	32.37
	$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-ST (base)	27.63	32.75	32.52
		FAT-ST (big)	28.13	33.39	32.72
	$D_{s,x,y} \cup D_{s,x} \cup D_{x,y}$	FAT-ST (base)	27.89	32.96	32.43
		FAT-ST (big)	28.80	34.28	34.22

Figure 4. Comparisons of the auxiliary MT task between MT baselines and our proposed methods

Table 5 shows the translation quality of auxiliary MT task of FAT-ST. Although our models trained with Must-C are worse than the MT baseline, by using FAT-MLM trained with more data, our proposed methods can easily outperform the MT baseline

Decoding Speed

Our proposed FAT-ST model is almost 2 faster than the Cascade system which needs to wait for the speech recognition module to finish before starting to translate. The decoding time of FAT-ST (big) is almost the same as FAT-ST (base) because we only increase the feedforward network in Transformers